

VAMSHI ADI

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EDUCATION

M.S. in Data Science | University at Buffalo

2021 - 2022

B.Tech in Computer Science | BVRIT

2011 - 2015

TECHNICAL SKILLS

Programming Languages: Go, Java, Python, JavaScript

Frameworks & Libraries: React, React Native, Node.js, Express, Django, TestNG, Cucumber

Distributed Systems & Net-TCP/IP, Load Balancing, Reverse Proxy, Service Discovery, Consul working:

Storage & Data Systems: MySQL, MongoDB, Cassandra, DynamoDB, Redis, ZFS, NFS

Cloud & DevOps: AWS (Lambda, API Gateway), Docker, CI/CD, Observability (Prometheus), Cloudflare

Security & Infrastructure: IAM, VPN (WireGuard/Tailscale), TLS, Firewall Policies, CrowdSec

EXPERIENCE

American Express

April 2024 – Present

Senior Software Engineer

Phoenix, USA

- Led development of **TestX**, a unified automation framework using Cucumber and TestNG, enabling parallel, tag-driven execution across multiple payment systems (AUTH, EDC, MultiRail, Token, Safekey, Fusion).
- Implemented trace-driven validation workflows with multi-hop (Hop2/Hop4) assertions and retry mechanisms, reducing regression cycles to **<30 minutes**.
- Architected **Cognix**, a high-performance TCP load balancer in Go, supporting low-latency, high-throughput transaction processing under **high concurrent load**.
- Contributed to modernizing a mainframe-based authorization network using **Java and Vert.x**, improving performance, reliability, and traceability.

TechnoVal Alliance (Client: Vitel Global)

March 2023 – April 2024

Senior Software Engineer

Phoenix, USA

- Engineered backend services in **Go** using an AWS serverless stack (Lambda, DynamoDB) for scalable, event-driven processing.
- Designed and implemented **custom authorization services**, improving security posture and streamlining access control workflows.
- Conducted feasibility analysis for migrating a serverless backend to **GCP**, evaluating scalability and performance tradeoffs.

Borderfree Technology

April 2020 – August 2021

Lead Software Engineer

Hyderabad, India

- Led a team of 6 engineers to design and deliver a scalable B2B portal using ReactJS and Go in a serverless architecture, improving system scalability and deployment efficiency.
- Architected REST APIs and custom authorization workflows using AWS Lambda.
- Built and launched React Native mobile applications across iOS and Android.
- Evaluated and prototyped GraphQL and gRPC integrations for backend systems.

NewsPlus

October 2018 – March 2020

Senior Software Engineer

Hyderabad, India

- Built a real-time news clustering system using Python microservices, powering feed generation, ranking, and notification systems at scale.
- Improved user engagement by **400%** by enabling personalized content delivery through clustering and recommendation pipelines.
- Implemented ML pipelines using K-Means, DBSCAN, and NLP techniques for scalable content processing.
- Integrated Firebase-based notification delivery across platforms, reducing latency and improving reliability.

Kony Inc.

June 2015 – September 2018

Senior Software Engineer

Hyderabad, India

- Built authentication systems including OAuth 2.0 and multi-session token management.
- Developed REST-based SDKs and cross-cloud file transfer frameworks for enterprise systems.

PROJECTS

Warforged Homelab Platform | *Docker, Traefik, Cloudflare, Linux, Networking*

- Designed and operated a self-hosted platform running multiple production-grade services with secure ingress via reverse proxy, TLS termination, and DNS management using Cloudflare.
- Engineered network isolation and secure service access using VPN routing (WireGuard/Tailscale) and segmented container networks.
- Architected distributed storage supporting **40TB+** of data using NFS over ZFS-backed systems, optimizing durability and throughput for high-bandwidth workloads.
- Diagnosed and resolved real-world performance bottlenecks impacting high-bitrate streaming by analyzing network throughput, container networking, and disk I/O constraints.
- Orchestrated multi-service environments with Docker, enabling reliable service discovery, routing, and lifecycle management across interconnected applications.
- Implemented layered security controls including CrowdSec, firewall policies, and identity-aware access patterns to reduce attack surface and enforce secure access.

Smart Wheelchair with Patient Monitoring | *IoT, Embedded Systems, Mobile Integration*

- Built a smartphone-controlled wheelchair integrated with real-time vital monitoring and alerting systems.
- Selected among top 10 projects for seed funding under Microsoft YouthSpark.

Campus Card - RFID Access & Payment System | *RFID, Distributed Systems*

- Designed an RFID-based system for access control and transaction tracking across institutional environments.
- Enabled real-time revenue tracking and multi-node system coordination.